

STAT X12: Binary Logistic Regression Practice

Part I

Donner Party example The theoretical model we outlined in the course notes can be estimated in R using the following code. Work with your groups or your table to use the output to answer the following questions.

```
options(show.signif.stars = FALSE)
library(Sleuth3)
library(tidyverse)
# rename dataframe
donner_df <- case2001
glimpse(donner_df)
```

```
Rows: 45
Columns: 3
$ Age      <int> 23, 40, 40, 30, 28, 40, 45, 62, 65, 45, 25, 28, 28, 23, 22, 23, ~
$ Sex      <fct> Male, Female, Male, Male, Male, Male, Female, Male, Male, Femal~
$ Status   <fct> Died, Survived, Survived, Died, Died, Died, Died, Died, Died, D~
```

```
# create new variables to match text
df_donner <- donner_df |>
  mutate(fem = factor(ifelse(Sex == "Female", 1, 0)),
         surv = factor(ifelse(Status == "Survived", 1, 0)))
# check new vars
glimpse(df_donner)
```

```
Rows: 45
Columns: 5
$ Age      <int> 23, 40, 40, 30, 28, 40, 45, 62, 65, 45, 25, 28, 28, 23, 22, 23, ~
$ Sex      <fct> Male, Female, Male, Male, Male, Male, Female, Male, Male, Femal~
$ Status   <fct> Died, Survived, Survived, Died, Died, Died, Died, Died, Died, D~
$ fem      <fct> 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 0, 1, ~
$ surv     <fct> 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, ~
```

```
# fit binary logistic regression model
fit_donner <- glm(surv ~ Age + fem,
                  data = df_donner, family = "binomial")
summary(fit_donner)
```

Call:

```
glm(formula = surv ~ Age + fem, family = "binomial", data = df_donner)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.63312	1.11018	1.471	0.1413
Age	-0.07820	0.03728	-2.097	0.0359
fem1	1.59729	0.75547	2.114	0.0345

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 61.827 on 44 degrees of freedom
Residual deviance: 51.256 on 42 degrees of freedom
AIC: 57.256

Number of Fisher Scoring iterations: 4

1. Write out the estimated model.
2. Interpret each of the estimated coefficients $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ on the *transformed (logit) scale*.
3. Interpret each of the estimated coefficients $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ on the *odds scale*.
4. Make a sketch of the “effects” plot you would expect to see from this model on the *transformed/logit scale*. Make another sketch on the *probability scale*.
5. Using what you know from reading the text, how do you expect other parts of the output to be used in writing a summary of statistical findings? Discuss this with your group/table, and then draft a summary of statistical findings.
6. Run the following code to check your work in 4.

```
# library effects package
library(effects)
# "effects" plot on the tranformed scale
plot(allEffects(fit_donner))

# on the original scale
library(ggeffects)
plot(ggpredict(fit_donner, terms = c("Age", "fem")),
     show_data = TRUE)
# using the effects package
plot(allEffects(fit_donner), type = "response")
```

Part II

Is the relationship between age and the odds of survival different for females than for males?

1. Write a complete theoretical binary logistic regression model that allows you to address the research question above. (Hint: The estimated model output is provided below).

```
# fit the interaction (full) model
fit_dint <- glm(surv ~ fem*Age, data = df_donner,
               family = "binomial")
# Wald-based tests are provided in summary output for
# testing beta_j = 0 for just one beta. Which beta is assoc with the interaction?
summary(fit_dint)
```

Call:

```
glm(formula = surv ~ fem * Age, family = "binomial", data = df_donner)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.31834	1.13103	0.281	0.7784
fem1	6.92805	3.39887	2.038	0.0415
Age	-0.03248	0.03527	-0.921	0.3571
fem1:Age	-0.16160	0.09426	-1.714	0.0865

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 61.827 on 44 degrees of freedom
Residual deviance: 47.346 on 41 degrees of freedom
AIC: 55.346

Number of Fisher Scoring iterations: 5

```
qnorm(p = 0.975)
```

```
[1] 1.959964
```

2. Make a sketch of the “effects” plot you would expect to see from `fit_dint` on the *transformed/logit scale*. Make another sketch on the *probability scale*.
3. Use Wald-based tests to investigate an interaction between age and sex on survival. Write a summary of statistical findings.

4. Regardless of your answer to 1, calculate a Wald-based 95% CI for β_3 (the interaction term).
5. Interpret the interval in 2 on the odds scale (i.e., back-transform and then interpret!).

The drop in deviance test approach for addressing the research question and the profile confidence intervals are provided below.

```
# LRT is
anova(fit_donner, fit_dint)
```

Analysis of Deviance Table

```
Model 1: surv ~ Age + fem
Model 2: surv ~ fem * Age
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      42      51.256
2      41      47.346  1   3.9099   0.048
```

```
confint(fit_dint)
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	-1.8113909	2.802618341
fem1	1.4134795	15.925314742
Age	-0.1168745	0.029437259
fem1:Age	-0.4026031	-0.001308492

```
exp(confint(fit_dint))
```

Waiting for profiling to be done...

	2.5 %	97.5 %
(Intercept)	0.1634267	1.648776e+01
fem1	4.1102320	8.246626e+06
Age	0.8896968	1.029875e+00
fem1:Age	0.6685774	9.986924e-01

6. Write a summary of statistical findings for the drop in deviance test investigating the RQ. How do the results compare to the Wald-based test?

7. Interpret the profile-likelihood-based 95% CI for β_3 on the odds scale. How does the interval compare to the Wald-based interval?
8. Refer back to 20.3 and 20.4 in your book. Discuss differences between the two inferential approaches with your group (Wald-based, LRT). Does the text provide any practical guidance for when to use which?
9. The empirical logit plot indicated that there may be some curvature in the overall association between age and survival. Outline an analysis plan for informally testing for a squared age term and an interaction between age and sex. With each step provide code you would use and provide the correct name of the statistical method.